

OSSP (Operating Systems And Systems Programming)

25CS2104E

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Section	8A
Practical	2

PRACTICAL-2

Aim: Develop a C program that uses the system calls **open()**, **read()**, **write()**, and **close()** to copy the contents of one file to another. Explain how control transitions between user space and kernel space during execution.

Use the **strace** utility to trace all system calls generated by the command **cat sample.txt**. Analyze the sequence of system calls and identify the kernel services involved.

Source: Practical-2 reference provided by the student.

Procedure

1. Open the Ubuntu/WSL terminal and navigate to the practical directory.
2. Create the source file **sample.txt** containing test text.
3. Compile the C source file using **gcc Practical_2.c -o Practical_2**.
4. Execute the compiled program using **./Practical_2**.
5. The program opens the source file, reads its contents, writes them to **copy.txt**, and closes both files.
6. Verify the copied contents using **cat copy.txt**.

Student Execution Output

```
Practical_2.c
1  #include <stdio.h>
2  #include <fcntl.h>
3  #include <unistd.h>
4
5  int main()
6  {
7      int source, destination;
8      char buffer[1024];
9      ssize_t bytesRead;
10
11     source = open("sample.txt", O_RDONLY);
12
13     if (source == -1)
14     {
15         perror("Error opening sample.txt");
16         return 1;
17     }
18
19     destination = open("copy.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);
20
21     if (destination == -1)
22     {
23         perror("Error opening copy.txt");
24         close(source);
25         return 1;
26     }
27
28     while ((bytesRead = read(source, buffer, sizeof(buffer))) > 0)
29     {
30         write(destination, buffer, bytesRead);
31     }
32
33     close(source);
34     close(destination);
35
36     printf("File copied successfully.\n");
37 }
```

The submitted terminal screenshot shows successful creation of **sample.txt**, verification of its contents, execution of **./Practical_2**, and the message **File copied successfully**.

System Call Tracing Using strace

After verifying the file-copy program, the **strace** utility can be used to observe the system calls generated while executing **cat sample.txt**.

Commands

```
strace -o trace.txt cat sample.txt  
grep -E "open|read|write|close|execve|exit_group" trace.txt
```

Typical sequence and analysis

System Call	Purpose
execve()	Loads and starts the cat program.
openat()	Opens sample.txt and obtains a file descriptor.
read()	Reads data from the opened file.
write()	Writes the file data to the terminal.
close()	Closes the file descriptor after use.
exit_group()	Terminates the cat process.

The reference practical reports that the trace includes calls such as **openat()**, **read()**, **write()**, and **close()**. These demonstrate how the user-space program requests file and I/O services from the kernel.

Kernel Services Involved

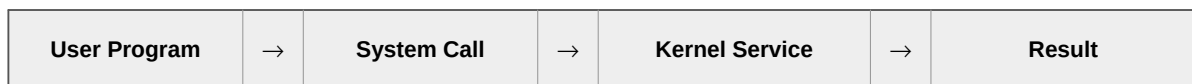
Service	System Calls	Role
Process Management	execve(), exit_group()	Process execution and termination.
File Management	openat(), close()	Opening files and managing file descriptors.
I/O Management	read(), write()	Transferring data between the process and kernel-managed I/O.

User Space and Kernel Space Transition

The C program executes initially in user space. When it invokes a system call such as **open()**, **read()**, **write()**, or **close()**, control transitions to kernel space. The operating system performs the requested protected operation and returns control and the result to the user-space process.

For this practical, the file-copy operation therefore demonstrates the interaction between a user-space application and kernel services for file access and I/O.

System Call Flow



Code

The following is the student's submitted **Practical_2.c** implementation.

```
vinay@DESKTOP-0CVM4SE:/mnt/c/Testing/OSSP_Testing$ echo "Hello World" > sample.txt
echo "This is my operating systems assignment." >> sample.txt
echo "This file will be copied using system calls." >> sample.txt
vinay@DESKTOP-0CVM4SE:/mnt/c/Testing/OSSP_Testing$ cat sample.txt
Hello World
This is my operating systems assignment.
This file will be copied using system calls.
vinay@DESKTOP-0CVM4SE:/mnt/c/Testing/OSSP_Testing$ ./Practical_2
File copied successfully.
vinay@DESKTOP-0CVM4SE:/mnt/c/Testing/OSSP_Testing$ |
```

Observation

1. The source file was successfully opened using **open()**.
2. File contents were read using **read()** and written to the destination file using **write()**.
3. **close()** successfully closed both files after copying.
4. During execution, control transitioned from user space to kernel space whenever a system call was invoked.
5. **strace** can trace the system calls generated by **cat sample.txt**.
6. System calls such as **openat()**, **read()**, **write()**, and **close()** can be observed in the trace.
7. The kernel provides services for file access, reading data, writing output, and closing file descriptors.

Result

The file was successfully copied using the **open()**, **read()**, **write()**, and **close()** system calls. The submitted execution produced the message **File copied successfully**. The **strace** procedure demonstrates how a user-space program interacts with kernel services through system calls.

Practical-2 Completed Successfully